

How a Golf Swing Carries Energy

A Visual Journey Through Timing, Forces, Models, and Uncertainty

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This is a story about a moving system. It follows energy from the ground, through the body and hands, into a flexible club, and finally toward the ball. The story uses ordinary scenes to make the mechanics visible, then

shows where those scenes stop being faithful. No swing instruction follows automatically from a model. What follows is an invitation to see the questions more clearly.

Follow the Transfer, Not Just the Speed Peaks



Each connection can carry power, store energy, dissipate energy, or redirect the load.

Figure 1: A Five-Stage Map Follows Mechanical Energy From the Ground Through the Body and Hands to the Club and Ball.

How to Read This Book

You can read straight through, or follow one of three paths:

- **The curious-reader path:** read the opening scene and the “Where the Picture Breaks” box in every chapter.
- **The mechanics path:** add the “Under the Hood” sections, which introduce the minimum mathematics needed to make each idea precise.
- **The evidence path:** follow “Go Deeper” links into the [complete technical paper](#), its source, and the primary literature.

Three labels recur throughout:

Model Result means that an executable mathematical system produced the result under declared assumptions. **Human Evidence** means people were measured. **Hypothesis** means a proposed explanation still needs a decisive test. These labels are intentionally difficult to overlook.

Chapter 1

Prologue: Follow the Energy

Imagine watching a relay race from high in the stands. You see one runner slow as the next accelerates. It is tempting to say that “speed was passed” from one runner to another. But speed is not an object. The baton is the object, chemical energy powers both runners, the track pushes on their shoes, and the exchange depends on where and when their hands meet.

A golf swing presents the same trap. Cameras show the pelvis peak first, then the torso, arm, and club. That order is important and widely observed in skilled golfers ([Putnam 1993](#); [Tinmark et al. 2010](#)). Yet a sequence of speed peaks does not by itself reveal the route taken by mechanical energy. To find the route, we need forces, moments, motion, and time.

This book therefore follows a different question:

When a distal segment becomes fast late in a linked motion, what carried, redirected, stored, supplied, or removed its energy along the way?

The answer will not be a slogan. Several pathways can operate at once. A joint can do direct work. A force at a moving connection can transport power. Gravity can add or remove energy. A shaft can temporarily store elastic energy. Damping can dissipate it. The visible motion is the combined account.

i The Theme

Think of the swing as a transport network whose roads change shape while the traffic is moving. The useful question is not merely “which town became busy first?” It is “what crossed each boundary, in which direction, at what rate, and with what loss or storage?”

1.1 Where the Picture Breaks

Energy is not a colored fluid traveling down the body. It is a bookkeeping quantity defined for a chosen system. A force does not carry a little packet that can be photographed. The transport-network picture helps us ask where to draw boundaries; the equations decide whether the account closes.

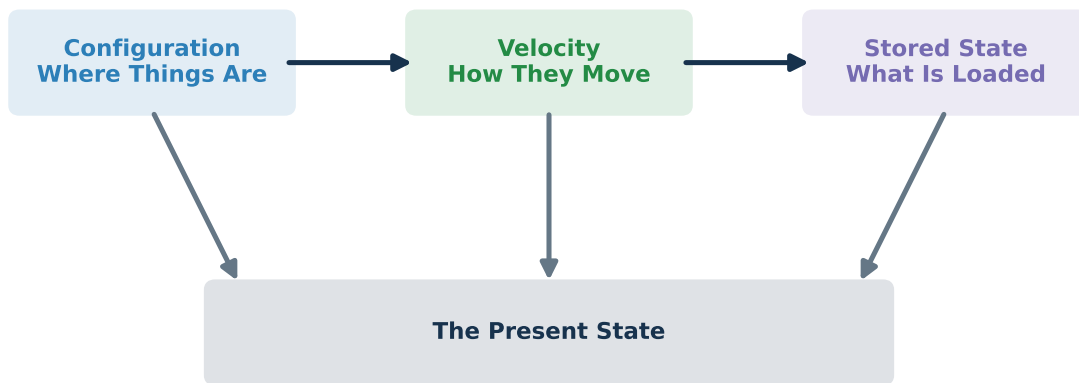
Go Deeper: The technical paper’s [mechanical-energy ledger](#) develops the formal accounting, drawing on segment-energy methods established by Robertson and Winter ([Robertson and Winter 1980](#)).

Chapter 2

A Swing Is a Moving System

Suppose someone freezes a swing for a photograph. The picture tells us where the body and club are, but not whether the club is rushing forward or drifting backward. Two identical photographs can be followed by very different futures.

A dynamics model needs a more complete snapshot called the **state**. At its simplest, the state contains configuration—where each part is—and velocity—how each part is moving. A flexible model also needs internal variables such as shaft deflection or muscle activation. Those variables remember loading that a single pose cannot show.



A model asks what happens next from this complete snapshot.

Figure 2.1: Three Boxes for Configuration, Velocity, and Stored Internal State Feed a Complete Present-State Snapshot.

2.1 The Recipe and the Kitchen

A model is like a recipe that must name its ingredients and rules:

1. **Parts:** masses, lengths, inertias, joints, and contact points.

2. **State:** configuration, velocities, and any internal storage variables.
3. **Inputs:** applied forces, torques, or activation commands.
4. **Environment:** gravity, ground contact, air, and the ball.
5. **Evolution rule:** equations that turn the present state and inputs into acceleration and a new state.

Changing any one of these can change the answer. A fixed shoulder, a moving base, and a full-body golfer are different systems—not higher and lower resolutions of one automatic truth.

2.2 Under the Hood: Why Acceleration Is Not Chosen Directly

For a linked mechanical system, a compact form is

$$M(q)\ddot{q} + h(q, \dot{q}) = Bu + J(q)^T \lambda.$$

Read it from left to right. The mass matrix M describes how difficult the current geometry is to accelerate. The term h groups velocity-dependent effects and gravity. The input u holds declared actuators. The constraint force λ enforces contacts or connections through the Jacobian J . Acceleration emerges only after those terms are reconciled.

! Where the Picture Breaks

A model is not a miniature person. It is a deliberately incomplete argument. Adding detail can answer a previously excluded question, but it can also add unknown parameters. More parts do not guarantee more knowledge.

Go Deeper: See the technical treatment’s [model hierarchy and common observables](#) and Featherstone’s reference on rigid-body dynamics ([Featherstone 2008](#)).

Chapter 3

Speed Is Not Energy

A light bicycle and a loaded truck can pass a sign at the same speed. Their speedometers agree; their kinetic energies do not. Energy depends on both motion and the properties of what is moving.

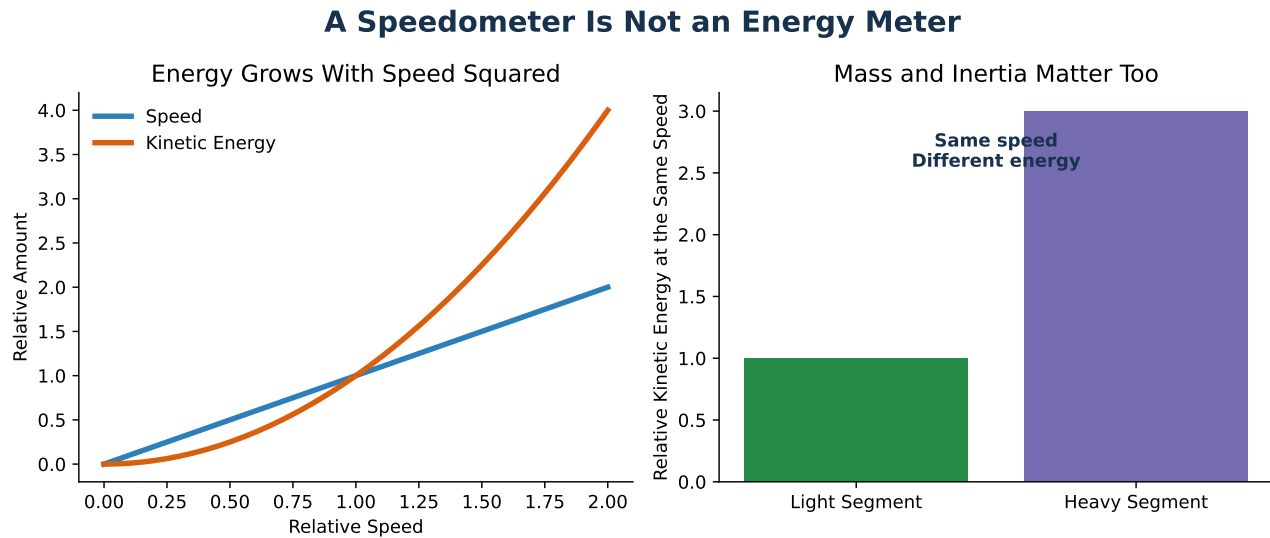


Figure 3.1: Two Plots Show That Kinetic Energy Grows With Speed Squared and That Objects at the Same Speed Can Carry Different Energy.

For a translating object, kinetic energy is $\frac{1}{2}mv^2$. For rotation, the parallel expression is $\frac{1}{2}I\omega^2$, where I is rotational inertia and ω is angular speed. In a linked body, cross-terms also matter because one segment carries another segment's axis through space.

This distinction explains why a kinematic sequence is suggestive but not a complete transfer measurement. A segment can slow while still receiving positive power if its inertia or geometry is changing. Another can speed up while an actuator removes energy, because interaction forces contribute even more positive power.

3.1 Power Is the Rate of the Exchange

Energy tells us how much is in an account. **Power** tells us how quickly the balance changes. For a force at a moving point,

$$P = F \cdot v.$$

Only the part of force aligned with velocity contributes instantaneous power. For a torque and angular velocity, $P = \tau\omega$. Power can be positive, negative, or zero even when force and torque are large.

Human Evidence: Segmental kinetic-energy analyses have found ordered outward energy patterns in golf, but the visible order and the energetic route remain distinct measurements ([Nesbit and Serrano 2005](#); [Kenny et al. 2008](#)).

! Where the Picture Breaks

The truck analogy treats mass as fixed and motion as simple translation. A golfer changes geometry, rotates in three dimensions, interacts with the ground, and deforms tissues and equipment. The analogy protects one lesson only: speed alone is not an energy audit.

Go Deeper: Read the technical paper's [distinction between kinematic sequence and energy transfer](#).

Chapter 4

The Club Is Carried Before It Is Released

Stand on a moving walkway while holding a suitcase. Before you swing the suitcase relative to your arm, the walkway already carries you, and your arm carries the suitcase. A distal object can acquire motion because its base is moving even when the distal joint is quiet.

The Distal Segment Is First Carried, Then Accelerated Relative to Its Base

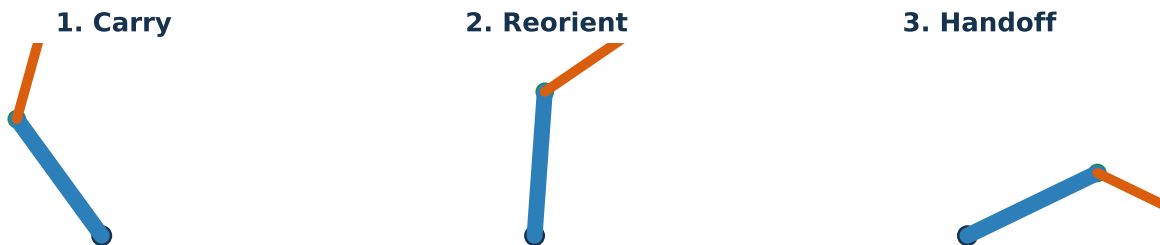


Figure 4.1: Three Linked-Segment Sketches Show the Club Being Carried, Reoriented, and Then Accelerated Relative to the Hands.

This is the heart of delayed distal handoff. Early in the downswing, proximal motion can carry the hands and club while the wrist angle remains comparatively retained. Later, the club rotates rapidly relative to the hands. That pattern does not mean the club was inert early or that all energy waited in one place. It means base motion and relative motion contributed differently over time.

Classic double-pendulum studies found that modeled clubhead speed can benefit when distal action is delayed rather than applied maximally from the top ([Milburn 1982](#); [Sprigings and Neal 2000](#); [Sprigings and MacKenzie 2002](#)). This does not make “delay” a universal instruction. The selected onset depends on the model, objective, limits, and starting state.

4.1 Why Early Drive Can Be Costly

Unfolding the distal link early can move its mass farther from the proximal rotation axis. That can increase the system's effective rotational inertia before proximal speed has developed. It also changes interaction forces and the direction in which those forces act. The mechanism is geometric, not mystical.

Model Result: In the baseline two-link comparison reported by the technical paper, immediate wrist drive produced less terminal clubhead speed than the passive-wrist case, while a selected later drive produced more. Those are finite-model comparisons, not estimates of how many metres any person will gain.

! Where the Picture Breaks

A whip tapers continuously and can carry traveling waves. A golf swing contains active joints, two hands, a moving base, and a comparatively stiff but flexible club. “Crack the whip” is therefore a poor mechanical identity. It can evoke sequencing, but it hides the actual pathways.

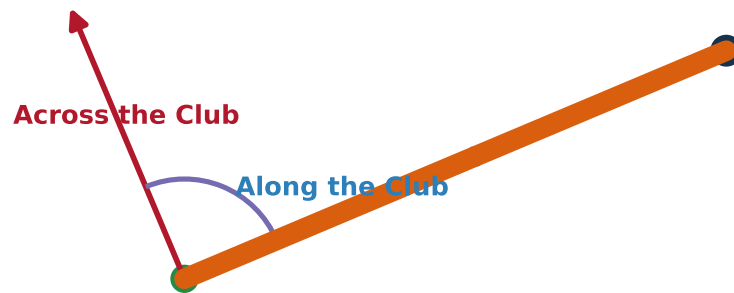
Go Deeper: Explore the paper's [92-command timing sweep](#) and the open double-pendulum literature ([Milburn 1982](#); [Spriggs and Neal 2000](#)).

Chapter 5

Forces Travel Through Connections

Push a shopping cart sideways while it rolls forward. Your force may turn the cart much more than it speeds it up. Push along the handle's path and more of your effort changes forward speed. Direction determines the consequence.

Direction Changes What a Force Can Do



The same force magnitude can pull, redirect, or turn depending on geometry.

Figure 5.1: A Club Diagram Resolves a Hand Force Into Components Along and Across the Club to Show How Geometry Changes Its Effect.

A joint connection can transmit force even if the joint actuator is doing no direct work. If the connection point moves, the force can transport power across the boundary. In segment-energy accounting, this is not a metaphor: $F \cdot v$ is the instantaneous boundary power.

The same force magnitude can therefore have different effects as the club changes orientation. Its tangential projection can accelerate rotation. Its radial projection can maintain curved motion. The moving hand path changes both projections throughout the swing.

5.1 Interaction Does Not Mean “Extra” Energy

Interaction forces redistribute and redirect energy within the linked system. They do not create energy from nothing. If the distal segment gains energy through a joint force, another account—proximal work, gravity, elastic storage, or existing kinetic energy—must support that gain, apart from numerical error.

Model Result: The open hand-path attribution studies found cases in which the pointwise drift contribution exceeded 100% of signed hand-path work because the simultaneous control contribution opposed it. Percentages above 100% can appear when signed terms cancel; they do not imply free energy.

! Where the Picture Breaks

“Force travels down the chain” can sound like a pulse in a pipe. Constraint forces are solved throughout the connected system at each state. Their values depend on the whole geometry, motion, loading, and chosen contact model.

Go Deeper: The [hand-path force attribution chapter](#) reports the complete force, impulse, power, and work decomposition.

Chapter 6

Two Hands Can Make a Turning Effect

Turn a steering wheel. One hand can push while the other pulls. The two forces may nearly cancel as a net push yet still create a strong turning effect because they act at separated points. Mechanics calls this a **couple**.

Opposing Hand Forces Can Form a Couple Without a Net Push

Narrow Grip

Wider Separation

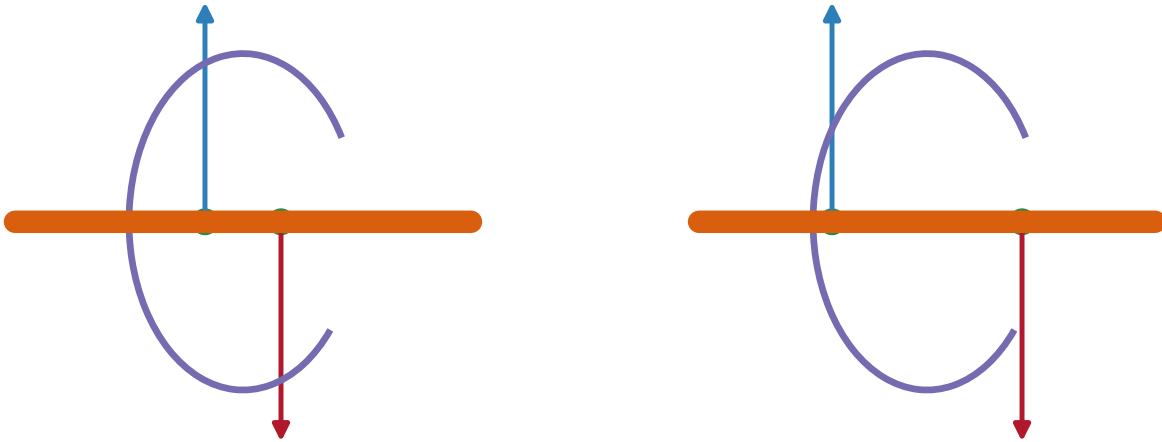


Figure 6.1: Two Grip Diagrams Compare Narrow and Wide Hand Separation Under Opposing Forces That Create a Turning Couple.

For two forces F_1 and F_2 applied at positions r_1 and r_2 , the moment about a declared reference point r_0 is

$$M_{r_0} = (r_1 - r_0) \times F_1 + (r_2 - r_0) \times F_2.$$

When the forces are equal and opposite, the net force can vanish while the couple remains. Increasing separation increases the moment for the same force geometry. Reversing the moment arms reverses the couple. Making the points coincident removes this particular mechanism.

Model Result: The reconstructed two-hand WSCG model and later forward constrained models produced a negative force-generated grip couple with no direct club torque in the declared branch. Coincident-grip controls collapsed that couple to zero, and reversed moment arms reversed its sign.

Human Evidence: Instrumented-grip research shows that individual hand forces can be measured and used in upper-limb kinetics ([Koike 2016](#); [Choi and Park 2020](#)). Those measurements are the appropriate bridge from a feasible mechanism to a claim about golfers.

! Where the Picture Breaks

The steering wheel is rigid and fixed to a simple axis. A golf club translates, rotates in three dimensions, bends, and receives forces from compliant hands. The couple picture explains separated-force geometry, not anatomy or intent.

Go Deeper: See the paper's [two-hand mechanism](#) and [zero-command controls](#).

Chapter 7

Negative Torque Is Not a Negative Intention

The word *negative* sounds evaluative. In mechanics it usually means only “opposite the chosen positive direction.” A negative torque is not bad, weak, braking, passive, or conscious by definition.

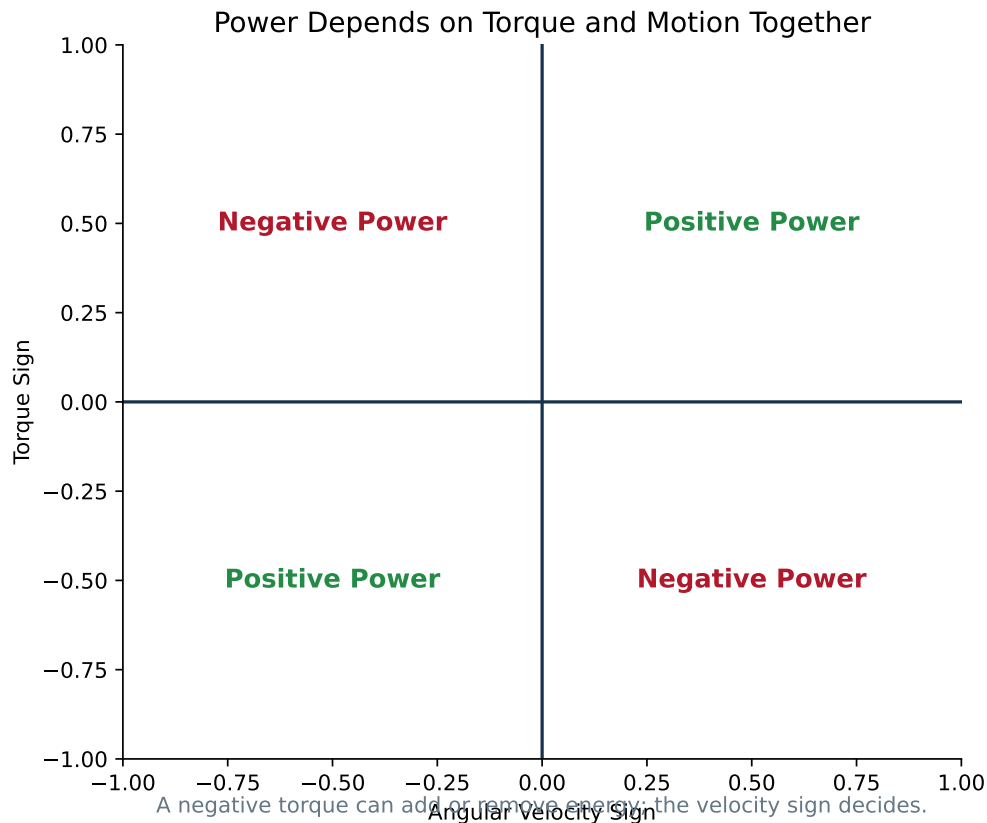


Figure 7.1: A Four-Quadrant Chart Shows That Mechanical Power Sign Depends Jointly on Torque Sign and Angular-Velocity Sign.

Torque alone does not determine energy flow. Power is $P = \tau\omega$. If torque and angular velocity share a sign, their product is positive. If their signs differ, power is negative. A negative torque applied during negative angular velocity supplies positive power.

Nor does a net joint torque uniquely identify muscle action. Many muscles can contribute, opposing muscles can co-contract, and contact forces can create a moment about the club without a direct wrist actuator. Inverse dynamics returns a net requirement under a model; it does not read the nervous system.

7.1 Passive Has Several Meanings

“Passive” might mean zero commanded torque, no positive work, no muscle activation, or motion produced without conscious intent. These are not equivalent. A zero-command model may still contain gravity, stiffness, damping, contact forces, and stored energy. A living wrist may generate a net moment through active co-contraction even if its angle changes little.

Scientific Boundary

The model establishes that negative torque and negative force-generated couples are mechanically possible under declared conditions. It does not establish which muscles produced them, whether a golfer intended them, or whether they are beneficial in a person. This is not a coaching instruction.

7.2 Where the Picture Breaks

A brake normally dissipates heat. Negative joint power can instead coexist with energy transfer through contact forces or with elastic storage elsewhere. Calling every negative torque a “brake” collapses distinct accounts.

Go Deeper: Read the technical paper’s [torque, couple, and power sign audit](#) and the foundational discussion of multi-articular muscle action ([Zajac and Gordon 1989](#)).

Chapter 8

A Shaft Can Store and Return Energy

Press a spring and release it. The spring returns some stored energy, but never chooses when to do so. Its release follows loading, stiffness, damping, and boundary motion. A golf shaft behaves similarly in principle, with bending and twisting distributed along its length.

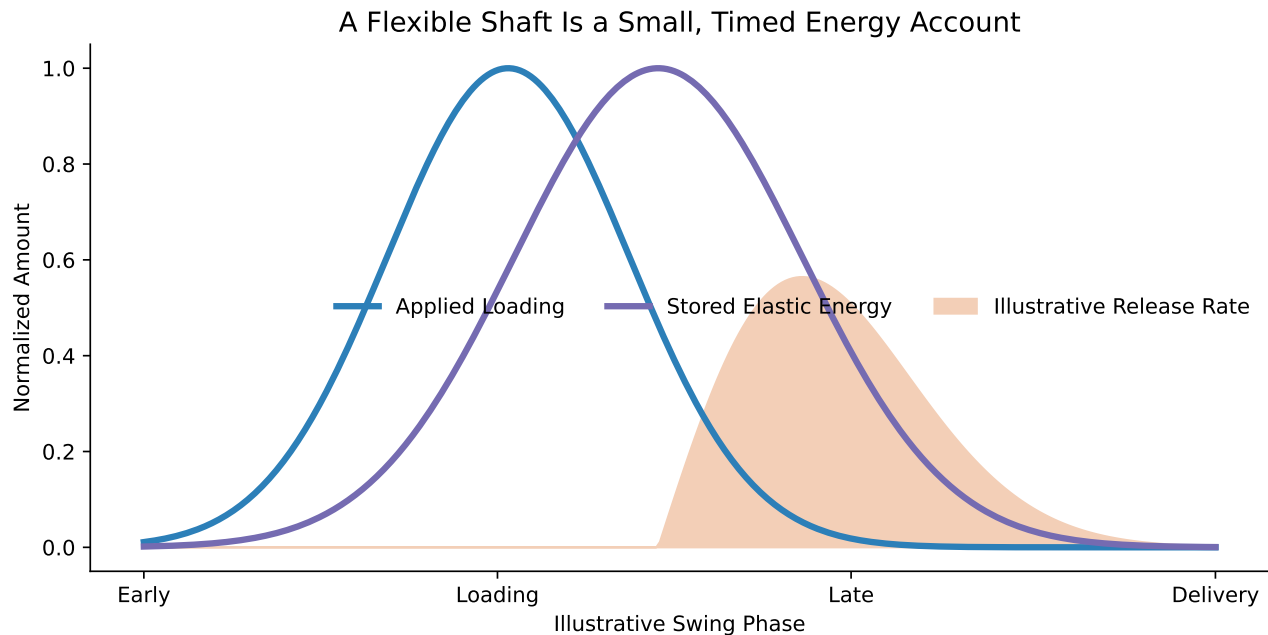


Figure 8.1: Three Illustrative Curves Show Applied Loading, Stored Elastic Energy, and a Later Release-Rate Window During the Swing.

The shaft is not a large hidden motor. It is a relatively small, timed account inside a much larger system. Its influence depends on how its modal response aligns with delivery. Fast load changes can excite modes that a one-mode model misses; slow changes may be represented more faithfully.

Model Result: In the matched rigid/flexible reference case, flexibility changed delivery speed modestly compared with gravity and damping ablations. The distributed-shaft study also showed much larger reduced-model error under a short pulse than a slow one. These findings argue for frequency-aware modeling, not a universal shaft benefit.

8.1 Under the Hood: Storage Is State

For a simple elastic coordinate s , stored energy is approximately $\frac{1}{2}ks^2$. Its rate depends on both deflection and deflection velocity. That makes preparation history observable: two identical club poses can have different futures if their internal shaft states differ.

! Where the Picture Breaks

A single coil spring has one deformation coordinate. A real shaft is a tapered, distributed structure with bending, torsion, damping, and clubhead boundary conditions. The spring picture explains storage but not the full spectrum.

Go Deeper: See the technical [distributed-shaft verification study](#).

Chapter 9

Timing by the Clock and Timing by the State

Traffic lights usually change by schedule or sensors. A scheduled light turns green after a fixed time. A sensor-controlled light waits until a vehicle arrives. When traffic speed varies, the sensor keeps the event tied to the actual situation rather than the clock.

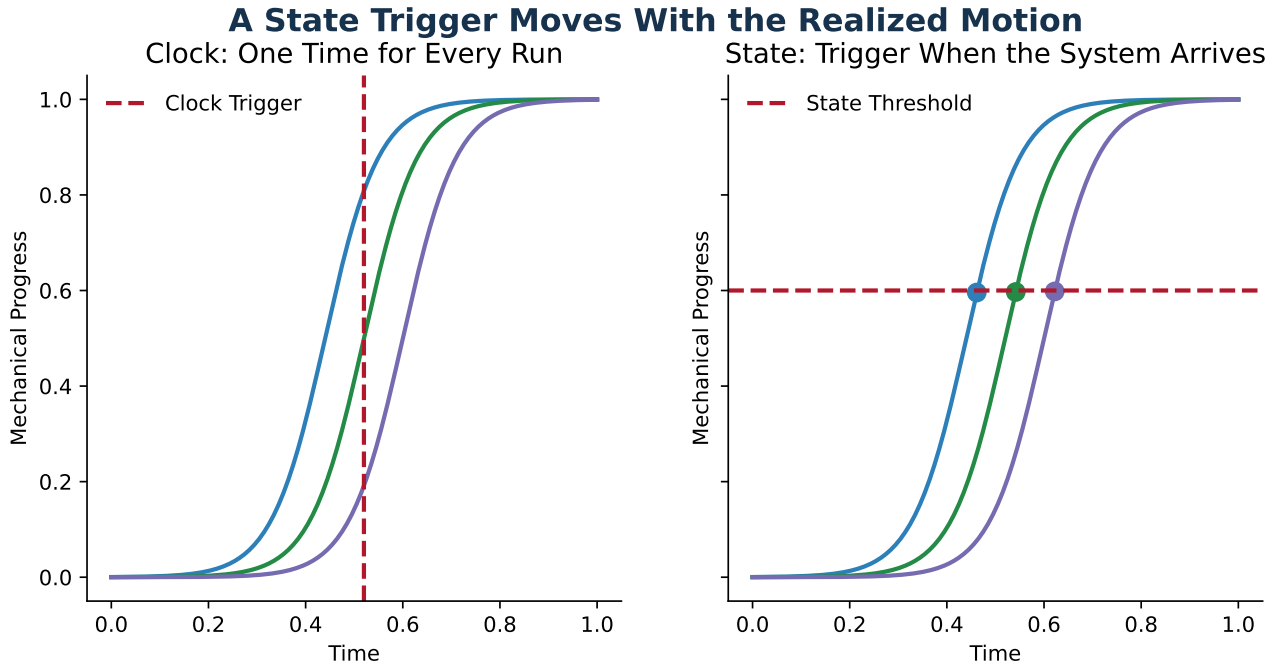


Figure 9.1: Two Plots Compare a Fixed Clock Trigger With a State Threshold That Moves in Time as the Realized Motion Changes.

A swing controller can be described the same way. A **clock trigger** changes the command at a fixed time. A **state trigger** changes it when a measured configuration or phase crosses a threshold. If the motion runs early or late, state triggering can follow the realized trajectory.

But sensing is not free. Real state feedback has noise, delay, ambiguity, and limited observability. A simulation with perfect state access is an upper bound on what that trigger could accomplish.

Model Result: In fifteen held-out perturbations, the state-triggered program raised the modeled tenth-percentile

There Is No Single Winner When the Outcomes Disagree

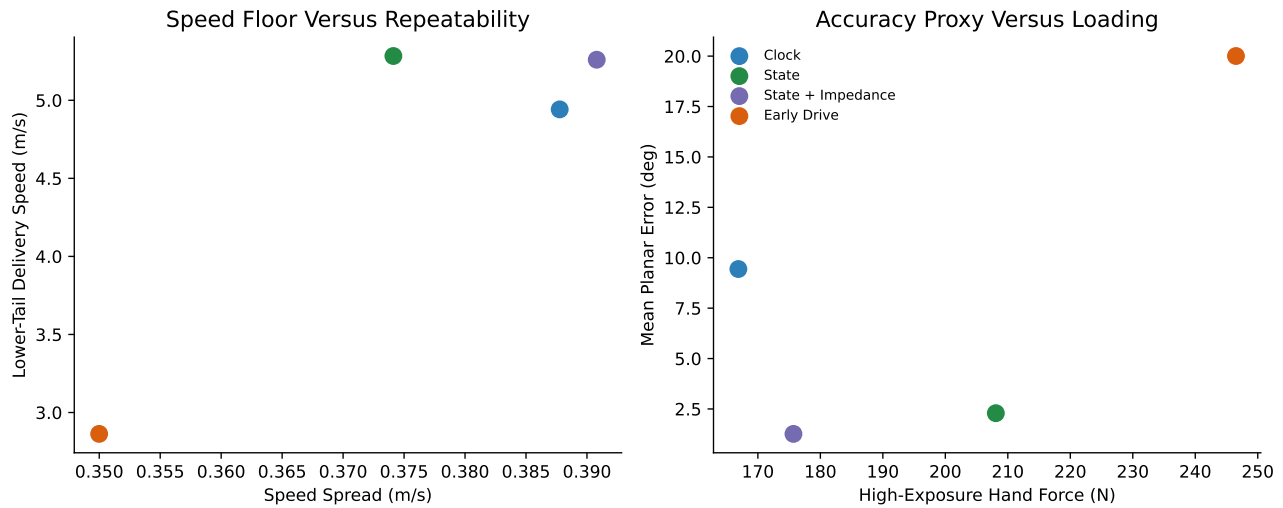


Figure 9.2: Two Evidence-Derived Scatter Plots Show the Tradeoffs Among Speed Floor, Speed Spread, Planar Error, and Hand-Force Exposure.

delivery speed from 4.94 to 5.28 m/s and reduced mean planar face/path error from 9.44° to 2.29° relative to the clock program. Its ninetieth-percentile peak hand force rose from 166.9 to 208.1 N. All four registered programs remained Pareto-nondominated: none was best on every objective. The complete numbers are in the [machine-readable evidence](#).

! Where the Picture Breaks

A traffic sensor observes a simple event. Human movement estimates state from vision, vestibular signals, proprioception, touch, and prediction, all with delay. “Use state feedback” is a research design, not a simple cue.

Go Deeper: Read [Transmission Pathways](#), [Robust Speed](#), and [Task Stability](#).

Chapter 10

Fast Once or Fast Repeatedly

Suppose two archers each fire ten arrows. One hits the bullseye once and scatters the rest. The other forms a tight group slightly off-center. “Best shot” and “best system” now mean different things.

Performance has at least three layers:

- **Central tendency:** the typical speed or outcome.
- **Dispersion:** how widely trials vary.
- **Tail behavior:** how bad the weak trials are, or how costly the extreme loads become.

Optimizing only nominal club speed can choose a fragile motion. Minimizing all movement variability can also be misguided, because some variation may occur in directions that barely affect the task.

Human Evidence: Golf movement variability does not map one-to-one onto outcome variability ([Tucker et al. 2014](#)). Launch-monitor variables also have different within- and between-session reliability, so observed dispersion must be separated from measurement noise ([Shaw et al. 2023](#)).

10.1 Robustness Means Naming the Disturbance

A motion is not robust in the abstract. It is robust *to something*: timing error, a changed initial pose, command scaling, shaft stiffness, grip geometry, or sensory delay. It must also be robust *for something*: speed, strike, orientation, loading, or injury-relevant exposure.

! Where the Picture Breaks

The archer can see the final target. A golfer’s delivery outcome is multidimensional: speed, face, path, strike location, dynamic loft, attack angle, and load all matter. A tight cluster in one coordinate may conceal a poor or unsafe tradeoff elsewhere.

Go Deeper: The technical paper’s [paired held-out perturbation design](#) shows how identical disturbances can compare candidate strategies fairly.

Chapter 11

Many Motions Can Share One Outcome

Write your name twice. The letters remain recognizable although no two pen paths are microscopically identical. The task allows many successful motions. Biological systems often possess more adjustable variables than the task strictly requires—a property called **motor abundance**.

Different Coordination Histories

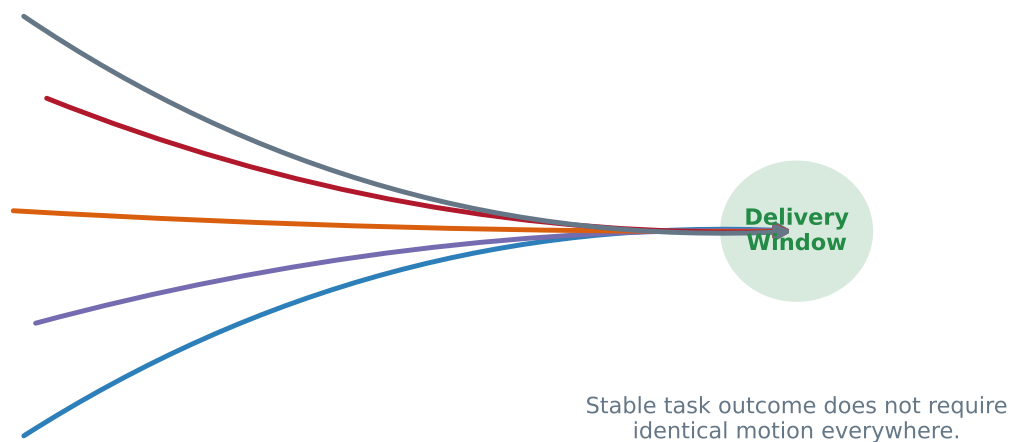


Figure 11.1: Five Differently Curved Coordination Paths Converge on the Same Green Delivery Window to Illustrate Task-Null Variability.

Near one operating point, we can ask how small input changes affect selected outcomes. The Jacobian J_y maps input perturbations δx to outcome changes δy :

$$\delta y \approx J_y \delta x.$$

Directions in the null space of J_y produce little first-order change in the chosen outcomes. They are not “wasted motion”; they may permit adaptation, comfort, or redistribution of internal load.

Model Result: The local eight-input, three-outcome map in the robustness study had rank three and nullity five. Most declared perturbation variance projected into the local task-null subspace. This is a property of that model, operating point, covariance, and outcome definition—not evidence of a neural synergy.

Hypothesis: Skilled golfers may organize variability so that delivery variables remain stable while less consequential coordinates vary. That idea requires participant-level repeated trials and nonlinear outcome analysis.

! Where the Picture Breaks

The signature analogy has a generous success criterion: humans can read many forms of the same name. Ball impact is less forgiving, and the relevant task manifold may curve sharply. A direction that is harmless infinitesimally can matter at larger amplitude.

Go Deeper: The technical chapter's [local task-null analysis](#) connects the model result to movement-variability literature ([Morrison et al. 2016](#)).

Chapter 12

What the Models Can and Cannot Tell Us

An architect's load calculation, a wind-tunnel model, a bridge prototype, and a decade of field records can all be excellent evidence. They answer different questions. A stronger equation does not automatically become field evidence; a larger dataset does not automatically identify a mechanism.

Confidence Rises Only When the Evidence Changes Kind

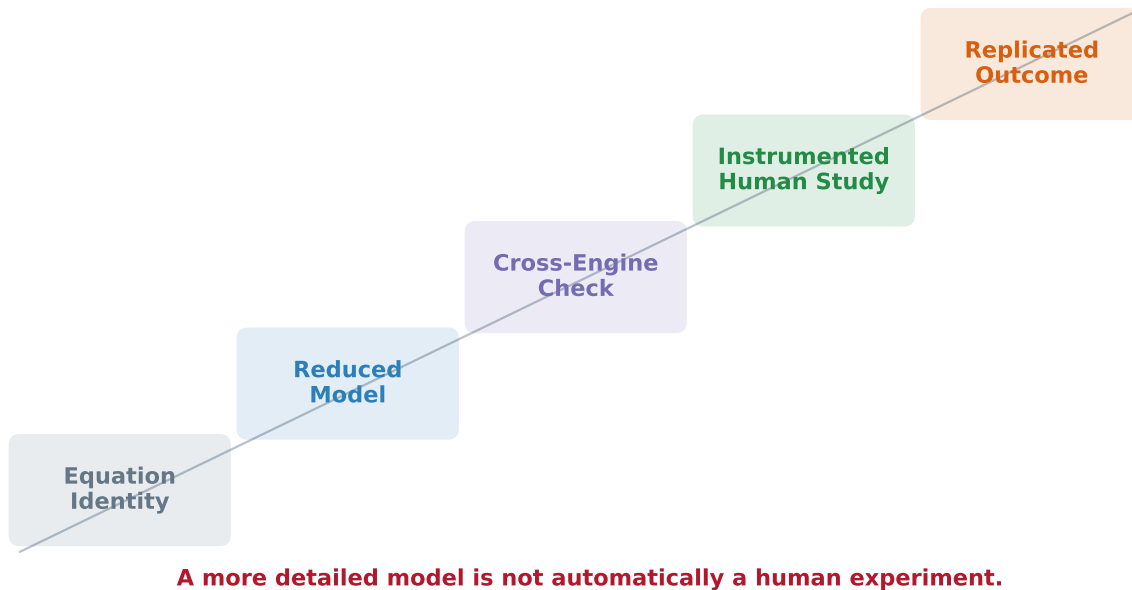


Figure 12.1: A Staircase Rises From Equation Identity Through Reduced Models and Cross-Engine Checks to Human Studies and Replicated Outcomes.

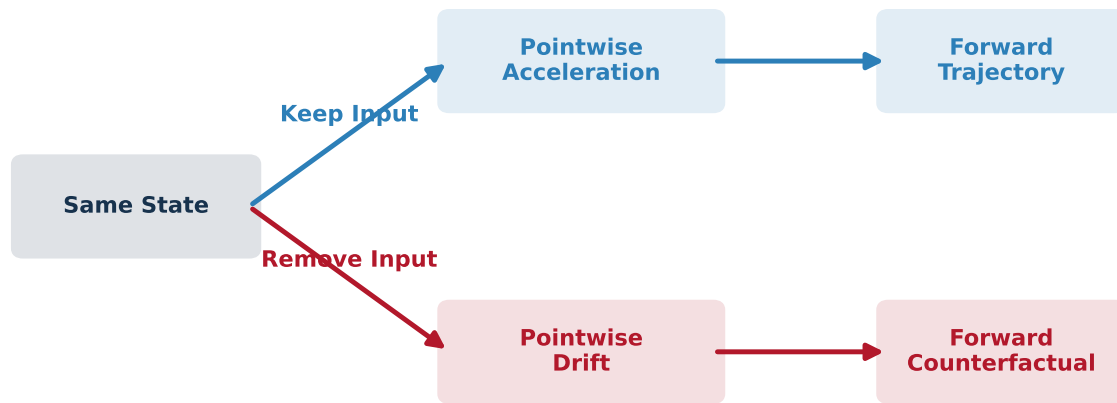
The project uses a ladder of model tiers:

1. A double pendulum isolates distal timing.
2. Prescribed two-hand models audit separated-force geometry.
3. Forward constrained models test whether the effect persists after inputs are removed.
4. Moving-base and flexible-shaft models admit additional pathways.
5. Spatial cross-engine studies test frame and implementation consistency.

6. A preregistered human protocol states what measurements could challenge the performance claims. Each rung should earn one new sentence, not permission to restate every earlier claim more confidently.

12.1 Counterfactuals Ask Two Different Questions

Two Questions That Must Not Be Confused



Pointwise: what is the acceleration now? Forward: where does the changed system go?

Figure 12.2: A Branching Diagram Distinguishes a Same-State Pointwise Acceleration Comparison From Forward-Integrated Counterfactual Trajectories.

A **pointwise counterfactual** asks: at the exact achieved state, what acceleration would the equations produce if a declared input were removed? A **forward counterfactual** branches from that state and integrates the altered system. The first attributes instantaneous acceleration. The second tests persistence. A stitched series of pointwise answers is not the same as one forward alternate history.

⚠ What Current Evidence Does Not Establish

The current package does not establish a universal swing optimum, a causal muscle strategy, passive human negative torque, injury reduction, or human self-stabilization. It does not convert a planar face/path proxy into ball flight. It does not identify individual-hand forces from a net wrench when the allocation is nonunique.

12.2 Where the Picture Breaks

An evidence ladder is not a single score. A clean algebraic identity may be more certain than a noisy human correlation, even though the human study is closer to application. Certainty and applicability are different axes.

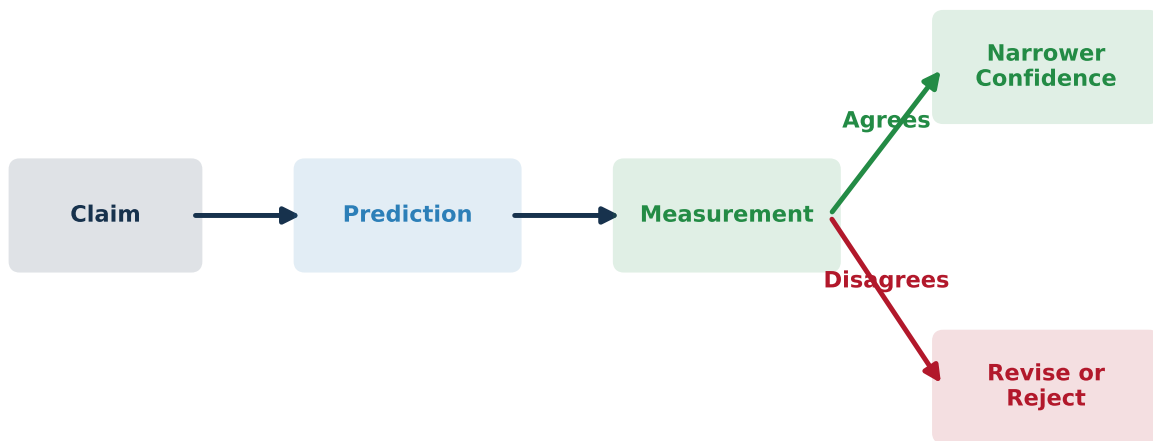
Go Deeper: Use the paper's [model-completion](#) and [falsification matrix](#) to see which claim belongs to which tier.

Chapter 13

A Research Program That Can Be Wrong

Good explanations take risks. They say what should happen, under which conditions, and what observation would count against them. A story that can explain every possible result after the fact has not explained enough.

A Scientific Story Must Include an Exit



A claim that survives every possible result has not risked enough.

Figure 13.1: A Claim Flows to a Prediction and Measurement, Then Branches Toward Narrower Confidence or Revision and Rejection.

13.1 Near-Term Tests

The strongest next experiment synchronizes whole-body motion capture, force plates, bilateral hand wrenches, shaft response, electromyography, club delivery, impact, and launch. It holds participants out of model selection, propagates measurement error, and perturbs phase rather than merely comparing self-selected swings.

The following observations **would count against** the proposed performance interpretation:

1. State-relative handoff fails to predict a better speed–dispersion–load frontier than clock-relative timing in participant-held-out data.
2. Measured bilateral forces cannot reproduce the sign or geometry of the proposed late grip couple after reference-point transport.
3. The apparent advantage disappears when strike, face, path, and ball flight replace terminal model speed.
4. The effect is smaller than instrument uncertainty or session-to-session variability.
5. Subject-calibrated models require direct distal actuation inconsistent with the proposed interaction-force pathway.

Failure would not erase Newtonian mechanics. It would narrow which mechanism matters in golfers and under which conditions. That is progress.

13.2 A Neutral Practical Reading

The current work suggests questions for measurement rather than universal cues:

- Does a transition follow the realized mechanical state or a fixed schedule?
- Which pathway supplies late club power: direct moment, moving contact force, elastic release, gravity, or a mixture?
- Does a higher-speed solution preserve face, path, strike, and acceptable loading across perturbations?
- Which movement variations are task-relevant, and which lie along an acceptable solution family?

These questions can organize experiments and individualized optimization. They are not a coaching instruction.

! Where the Picture Breaks

Science does not advance as a tidy staircase from ignorance to certainty. Measurements disagree, definitions evolve, models fail for informative reasons, and useful explanations retain explicit boundaries.

Go Deeper: The project’s full [adversarial review](#) lists each gap, counterexample, falsifier, and forward action.

Glossary

Acceleration: The rate at which velocity changes. It includes changes in speed and direction.

Constraint: A rule limiting possible motion, such as two modeled contact points remaining on a grip.

Control: A declared model input. It need not correspond one-to-one with a muscle, intention, or cue.

Couple: A turning effect produced by separated forces, often with little or no net force.

Distal / Proximal: Farther from / closer to the body’s central attachment along a linked chain.

Drift: The acceleration field remaining when declared control inputs are set to zero at a state. Drift can still include gravity, velocity-dependent terms, stiffness, damping, and constraints.

Energy: A conserved bookkeeping quantity for the ability of a declared system to produce change, partitioned into kinetic, potential, elastic, and other forms.

Impedance: The motion-dependent resistance created by inertia, stiffness, damping, and control. It is not synonymous with simply “being stiff.”

Inertia: Resistance to acceleration. Rotational inertia depends on how mass is distributed relative to an axis.

Kinematic Sequence: The observed order of segment angular-speed peaks. It is not, by itself, an energy-pathway measurement.

Momentum: A state quantity associated with mass and velocity; angular momentum is its rotational counterpart.

Pareto-Nondominated: No other tested candidate improves one objective without worsening at least one other objective.

Power: The instantaneous rate of energy transfer or work. Translational power is $F \cdot v$; rotational power is $\tau\omega$.

Reference Frame / Reference Point: The axes and point relative to which a vector or moment is expressed. Moments must be transported consistently when the point changes.

Robustness: Preservation of a declared outcome under declared disturbances.

State: The information required by a model to predict its next evolution, usually configuration, velocity, and relevant internal variables.

Task-Null Direction: A local input change that produces little first-order change in selected task outcomes.

Torque or Moment: A turning effect about an axis or point. Its sign alone does not determine power.

Work: Energy transferred by a force or torque through motion, accumulated over an interval.

Wrench: A force and moment expressed together at a declared point and in a declared frame.

ZTCF: Zero-torque counterfactual. The term must specify whether it is a pointwise same-state evaluation or a forward-integrated branch.

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